KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2023
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

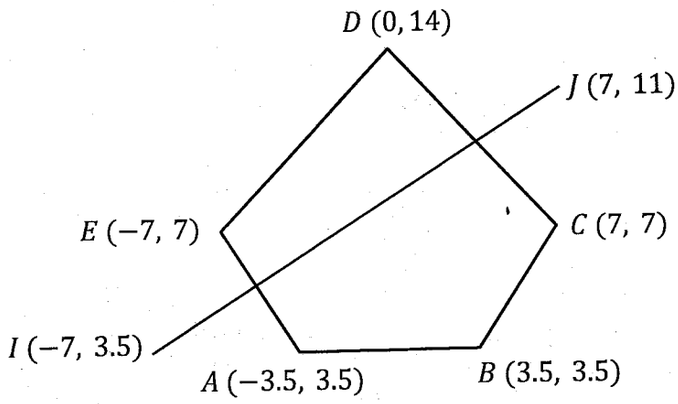
TIME: 3 hours | FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates the course outcomes.

SECTION A

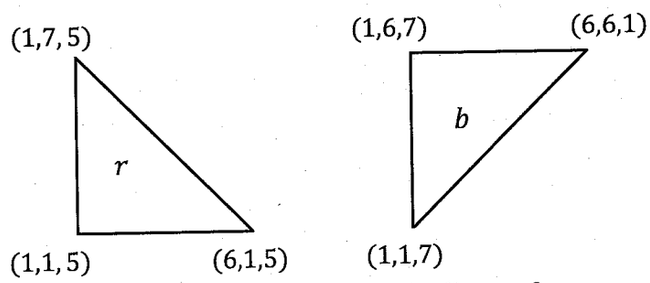
(Answer **ANY THREE** questions from this section in Script A)

1. a) Suppose you are given a task to choose a display system for your project. It needs to be able to draw millions of lines, triangles or characters on the screen. Which display system will you choose? Explain your reasonings. (08) [CO1]
   b) Describe the types of fractals that can be found in the real world. (09) [CO1]
   c) Figure below (not drawn to scale) shows a clipping polygon defined by five vertices. A line IJ is defined by two vertices I and J. Clip the line IJ against the clipping polygon using Cyrus-Beck algorithm and draw the final result after clipping. (18) [CO2]



$$t = \frac{N_i.[P_{E_i} - P_0]}{N_i.D}$$

2. a) What steps are required to scan convert an ellipse using the trigonometric method. (10) [CO2]
   b) The following figure shows two triangles and its vertices. Using Z-buffer, calculate the color of the pixels of color buffer. Show the depth buffer in each step. Assume that the color buffer has 8 (eight) rows and columns. The first triangle is red, denoted by *"r"* and the second one is blue, denoted by *"b"*. Write *"r"* or *"b"* to denote color red and blue, and *"bc"* for background color. (15) [CO2]



   c) Suppose, you are given a task of generating texture coordinates for an actual 3D model of earth. Describe two ways of achieving your goal. (10) [CO1]

3. a) Write an algorithm to determine if a ray intersects a triangle or not. Be sure that it is not the plane containing the triangle but the triangle itself. (12) [CO2]
   b) A line P_0P_1, described by two points $P_0 = (21.2, 12.3)$ and $P_1 = (28.8, 15)$, is given. You have to rasterize the line using Bresenham's line drawing algorithm. Calculate five consecutive points on the line starting from P_0 [excluding P_0]. (15) [CO2]
   c) "On a raster display, diagonal lines feel dimmer than a horizontal or vertical line." Do you agree with the statement? Explain your reasonings. (08) [CO1]

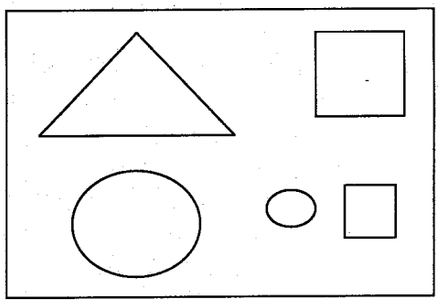
4. a) Describe how mipmapping achieves anti-aliasing. (09) [CO1]

b) Suppose there is a scene consisting of two opaque spheres. A red sphere with center $C_r = (2, 4, -10)$ and radius $R_r = 5$, and a blue sphere with center $C_b = (4, 8, -20)$ and radius $R_b = 3$. We want to calculate the color of a pixel which have direction vector $d = (0.18, 0.36, 0.91)$from the camera origin. Calculate the color of that pixel. For simplicity let's assume there is only white ambient light in the scene. The ray tracing tree shouldn't be deeper than two levels. The background color is black. (20) [CO3]

For your convenience here are some formulas, where the symbols have their usual meanings in this context.

$$t = \frac{-B \pm \sqrt{B^2 - 4C}}{2}; B = 2(s - c).d; C = |s - c|^2 - R^2; r = u - 2(u.n)n$$

c) The following figure shows a 2D slice of a 3D scene. Draw the approximate output of applying (i) KD tree (ii) Octree algorithm to divide the scene into multiple subspaces. (06) [CO2]



SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Derive the scaling matrix for scaling an object by a scale factor 3 along an arbitrary direction given by vector $u = (1, 2)$ rooted at $(4, 3)$. (08) [CO1]

b) Use quaternion basics to show that (13) [CO2]

$$\boldsymbol{qPq}^{-1} = (s^2 - \boldsymbol{v}^2)\boldsymbol{P} + 2s\boldsymbol{v} \times \boldsymbol{P} + 2(\boldsymbol{v}.\boldsymbol{P})\boldsymbol{v}$$

where **P** is a pure quaternion and $\boldsymbol{q} = [s + \boldsymbol{v}]$ is a unit quaternion.

c) The Camera Coordinate System (CCS) is defined by the basis vectors $\boldsymbol{u}(0.782, 0.569, 0.254)$, $\boldsymbol{v}(0.079, 0.927, -0.367)$ and $\boldsymbol{n}(0.334, 0.668, 0.665)$. Currently the camera is at world origin, and you sequentially pitched $(0^\circ \leq \theta^\circ \leq 90^\circ)$ and rolled $(-90^\circ \leq \beta^\circ \leq 0^\circ)$ the camera. (14) [CO3]

If after transform the basis vectors are $\boldsymbol{u}'(0.425, 0.828, -0.366)$, $\boldsymbol{v}'(-0.0044, 0.166, -0.986)$ and $\boldsymbol{n}'(-0.166, 0.955, -0.246)$, find the pitch, θ and roll, β. [Hint: Following equations are given for your convenience.]

$$a \sin x + b \cos x = c$$
$$x = \sin^{-1}(c/\sqrt{a^2 + b^2}) - \tan^{-1}(b/a) + 2n\pi$$
$$or, x = \pi - \sin^{-1}(c/\sqrt{a^2 + b^2}) - \tan^{-1}(b/a) + 2n\pi$$

6. a) Consider the following perspective matrix defined for projection plane containing point $R_0(x_0, y_0, z_0)$, plane normal $N(n_1, n_2, n_3)$, and projection reference point $P_{prp}(a, b, c)$, with $d = d_0 - d_1 = N.R_0 - N.P_{prp}$. (12) [CO3]

$$\begin{bmatrix} d + an_1 & an_2 & an_3 & -ad_0 \\ bn_1 & d + bn_2 & bn_3 & -bd_0 \\ cn_1 & cn_2 & d + cn_3 & -cd_0 \\ n_1 & n_2 & n_3 & -d_1 \end{bmatrix}$$

Show that, for this perspective transform, vanishing point P_v in the direction vector U can be formulated as $P_v = P_{prp} + \left(\frac{d}{K}\right) U$ where $K = N.U$. Show the derivation steps in details.

b) Explain the importance of using Distribution Term, D and Geometry Term, G in their illumination model by Cook-Torrance. How do they estimate those terms? Show with necessary illustration, reasoning and derivation. (15) [CO1]

c) For the given control points (0,0), (0,1), (4,0), (5,0); use Hermite blending functions to generate a cubic curve $Q(t)$. Find the point on the curve at $t = 0.2$. (08) [CO2]

7. a) Let you are drawing a non-uniform cubic B-spline curve $Q(t)$, with control points $(5,6),(10,25),(13,10),(20,30)$ and $(26,15)$ and knot vector [0,0,0,0,2,4,5,6,7]. Determine the blending functions for the parameter range $2 \leq t \leq 4$. Also find the point on the curve $Q(t)$ at $t = 3.0$. For your convenience, the recursive Cox-deBoor blending equations are given below, where symbols have their usual meanings. (16) [CO1]

$$B_{k,1}(t) = \begin{cases} 1, & if\ t_k \leq t \leq t_{k+1} \\ 0, & Otherwise \end{cases}$$

$$B_{k,d}(t) = C_k B_{k,d-1}(t) + (1 - C_{k+1}) B_{k+1,d-1}(t)$$

$$C_k = \frac{t - t_k}{t_{k+d-1} - t_k}$$

b) Let we want to define a parametric surface patch $S(u,v)$ using Bezier blending functions. Suppose the patch is quadratic and cubic on the parametric directions u and v respectively. How many control points do you need to supply? Represent the patch in tensor notation, clearly stating each item. Particularly show the Bezier basis matrices for different directions by generating from Bernstein polynomial equation. (11) [CO3]

c) Let a plane P contains the origin and a unit vector **u** is orthogonal to the plane. A reflection across P is a linear transform that maps each point **x** to its mirror image directly across P. Prove that this reflection transform for such a plane can be represented by a 3×3 matrix $\boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^T$. Write out the matrix in component form too. (08) [CO2]
[Hint: Use idea similar to the rotation matrix derivation used by Rodrigues, where $v_{ref} = v_{\perp} - v_{\parallel}$]

8. a) Use Gauss-Seidal iterative approach of solving radiosity equation to find the radiosities of each of the patches having following information. Approximate values up to four decimal places. (20) [CO2]

Patch	1	2	3	4
Emissive Power	0	0	0.6	0.8
Area	0.4	0.6	0.5	0.2
Reflectivity	0.8	0.6	0.3	0.3

		To			
		1	2	3	4
From	1	–	0.5	0.6	0.3
	2	–	–	0.4	0.1
	3	–	–	–	0.2

Form factors

b) Explain why Gouraud surface rendering method produces much better color approximation than constant shading method, especially for curved surfaces represented by polygonal mesh. (07) [CO3]

c) Derive the equations for computing delta form factor of a pixel located at back face of a hemicube. (08) [CO1]